1.Description

The KIA1117 series of positive adjustable and fixed regulators is designed to provide 1A for applications requiring high efficiency.All internal circuitry is designed to operated down to 800mA input to output differential and the dropout voltage is fully specified as a function of load current.

The KIA1117 offers current limiting and thermal protection.The on chip trimming adjusts the reference voltage accuracy to 2%.

2.Features

- Output current of 1A typical
- Three-terminal adjustable or fixed
- 1.5V,1.8V,2.5V,2.85V,3.0V,3.3V,5.0V outputs
- Thermal protection built in
- Fast transient response

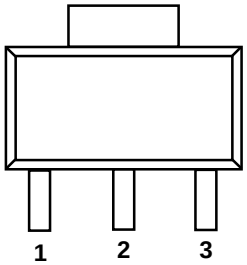
3.Applications

- PC motherboard
- LCD monitor
- DVD-Video player
- NIC/switch
- Telecom equipment
- High efficiency linear regulator
- Printer and other peripheral equipment
- Battery charger

4.Pinning information

Table1: Pinning-SOT223,simplified outline

Pin	Description	Simplified outline
1	ADJ/GND	
2	V_{OUT}	
3	V_{IN}	



(SOT223 Front View)

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5.Package information

2.5K/Reel 5K/Box 40K/CTN

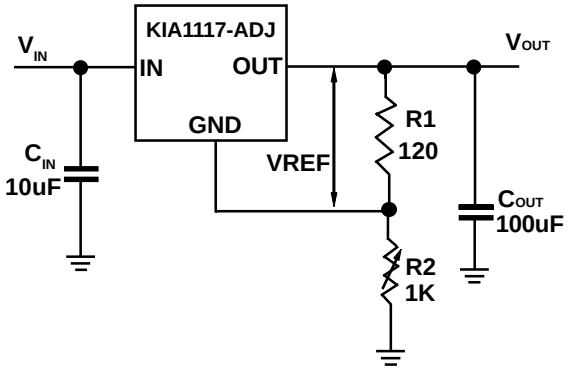
6.Maximum ratings(Ta=25℃)

Table2: Maximum ratings

Parameter	Symbol	Rating	Unit
Input voltage	V_{IN}	12	V
Operating junction temperature	T_J	0~+125	℃
Storage temperature range	T_{STG}	-40~150	℃

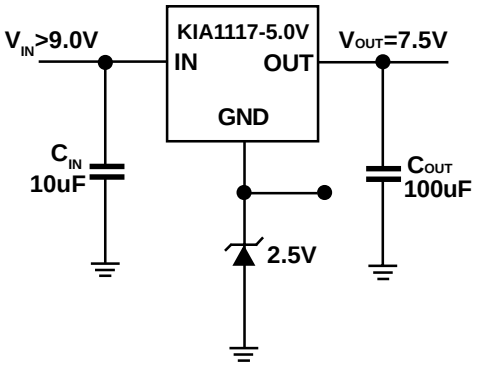
7.Typical application

Table3: Typical application



V_{OUT}= VRE F (1+ R2/R1)+IADJR2

Application 1:1.25V to 10V Adjustable Regulator



V_{OUT}=7.5V

Application 2:Regulator with reference

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8.Electrical characteristics

Table4: Electrical characteristics

(V_{IN}=10V,I_{OUT}=40mA,C_{IN}=0.33uF, C_{OUT}=0.1uF,T_J=25°C ,Unless otherwise noted)

Parameter		Symbol	Condition	Min	Typ	Max	Unit
Reference voltage		V_{REF}	I_{OUT}=10mA,V_{IN}-V_{OUT} =1.5V	1.237	1.250	1.263	V
			10mA≤I_{OUT}≤1A V_{OUT}+1.5V≤V_{IN}≤12V	1.225	1.250	1.275	V
Output voltage	1.5v	V_{OUT}	10mA≤I_{OUT}≤1A,3.0V≤V_{IN}≤12V	1.470	1.500	1.530	V
	1.8v	V_{OUT}	10mA≤I_{OUT}≤1A,3.3V≤V_{IN}≤12V	1.764	1.800	1.836	V
	2.5v	V_{OUT}	10mA≤I_{OUT}≤1A, 4.0V≤V_{IN}≤12V	2.450	2.500	2.550	V
	2.85v	V_{OUT}	10mA≤I_{OUT}≤1A,4.35V≤V_{IN}≤12V	2.793	2.850	2.907	V
	3.0v	V_{OUT}	10mA≤I_{OUT}≤1A,4.50V≤V_{IN}≤12V	2.940	3.000	3.060	V
	3.3v	V_{OUT}	10mA≤I_{OUT}≤1A,4.80V≤V_{IN}≤12V	3.234	3.300	3.366	V
	5v	V_{OUT}	10mA≤I_{OUT}≤1A,6.50V≤V_{IN}≤12V	4.900	5.000	5.100	V
Line regulation		$\triangle V_O$	I_{OUT}=10mA,V_{OUT}+1.5V≤V_{IN}≤12V		0.035	0.2	%
			I_{OUT}=10mA,V_{OUT}+1.5V≤V_{IN}≤12V		1.0	6.0	mV
Load regulation		$\triangle V_O$	10mA≤I_{OUT}≤1A, V_{IN}-V_{OUT} =3V		0.2	0.4	%
			10mA≤I_{OUT}≤1A,V_{IN}=V_{OUT} +1.5V		1.0	10.0	mV
Dropout voltage		$\triangle V$	I_{OUT}=10mA		0.8	1.15	V
			I_{OUT}=1A		1.1	1.30	V
Minimum load current(Note1)		I_{MIN}	V_{IN}≤12V		5		mA
Quiescent current		I_Q	V_{IN}≤12V		5	10	mA
Current limit		I_{CL}	V_{IN}-V_{OUT} =5V	0.9	1.1	1.5	A
Adjust pin current			I_{OUT}=10mA, V_{IN}-V_{OUT} =2V		50	120	uA
Thermal regulation (Note2)			T_A=25°C,30 ms pulse		0.008	0.04	%/W
Ripple rejection (Note2)		R_R	f_o=120Hz,1V_{RMS},I_o=400mA, V_{IN}-V_{OUT} =3V	60	75		dB

Note1:For the adjustable device,the minimum load current is the minimum current required to maintain regulation. Normally the current in the resistor divider used to set the output voltage is selected to meet the minimum load current equirement.

Note2: These parameters,although guaranteed,are not tested in production.

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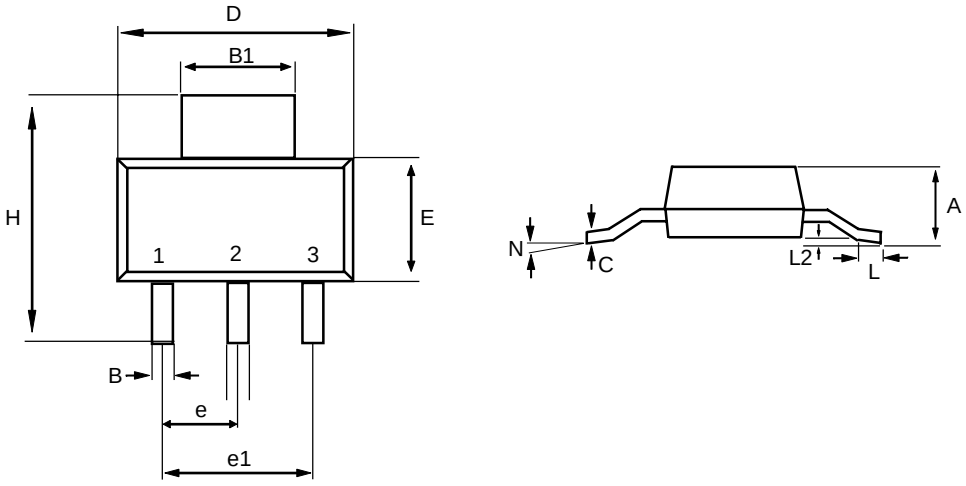
10.SOT223 package outline

Table5: SOT223 package outline

DIMENSIONS(mm are the original dimensions)



Dim	Min	Max	Dim	Min	Max
A		1.80REF	e		2.30REF
B	0.60	0.80	e1		4.60REF
B1	2.90	3.10	H	6.70	7.30
C	0.24	0.32	L	0.91	
D	6.30	6.70	L2	0.02	0.10
E	3.30	3.70	N	0^{o}	10^{o}

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